SERVERLESS CUSTOMER ANALYTICS PIPELINE

End-to-End Data Engineering Portfolio · ETL Automation · Cloud Architecture · NLP Sentiment Analysis

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AWS Lambda	Amazon S3	EventBridge	Pandas	BeautifulSoup4	PostgreSQL	Power BI

~$0	100%	Daily	3-Tier
Monthly Infrastructure Cost	Zero-Touch Automation	Pipeline Execution	Data Architecture

01 · PROJECT OVERVIEW

An end-to-end, fully automated ETL pipeline that converts raw public customer reviews into structured business intelligence — with zero manual intervention. Designed with a **Serverless-first** mindset, the system operates at near-zero cost while delivering daily-refreshed sentiment dashboards to decision-makers.

02 · PIPELINE ARCHITECTURE

01	EXTRACT	AWS EventBridge triggers Lambda daily. Python scrapes Trustpilot using Requests + BeautifulSoup4, collecting raw HTML reviews at scale.
02	TRANSFORM	Pandas cleans and structures raw data. A rule-based NLP model classifies each review as Positive / Neutral / Negative based on star rating.
03	LOAD · S3	Raw CSV is archived in Amazon S3 as the Data Lake layer — ensuring full traceability, reprocessing capability, and long-term backup.
04	LOAD · DB	Clean DataFrame is appended to a PostgreSQL warehouse (Supabase) via SQLAlchemy + pg8000, optimized for serverless connection pooling.
05	ANALYZE	Power BI connects via DirectQuery to the PostgreSQL warehouse, auto-refreshing dashboards with zero latency for business stakeholders.

03 · BUSINESS VALUE

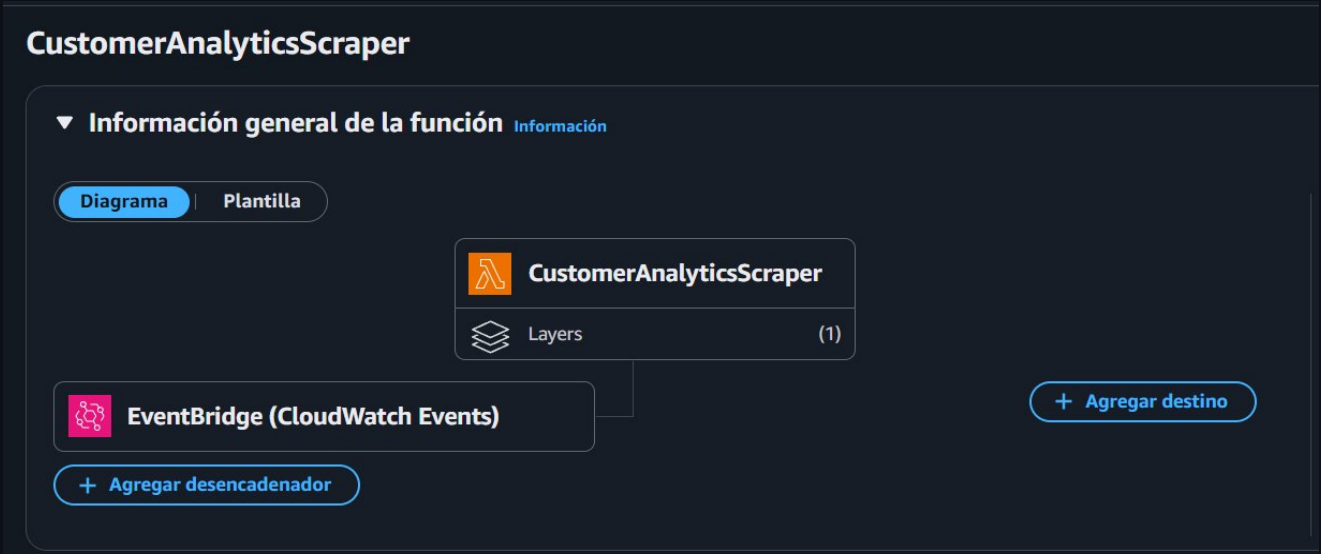
■ COST REDUCTION	■ AGILE DECISIONS	■ SCALABILITY

Serverless billing means the pipeline costs pennies per month — paying only per execution second, not for idle servers running 24/7.	Marketing & CX teams access a live Power BI dashboard updated every 24h, enabling instant response to PR crises or campaign performance shifts.	The model replicates effortlessly to monitor competitors, giving the business a continuous competitive intelligence edge at no extra infrastructure cost.

04 · TECH STACK

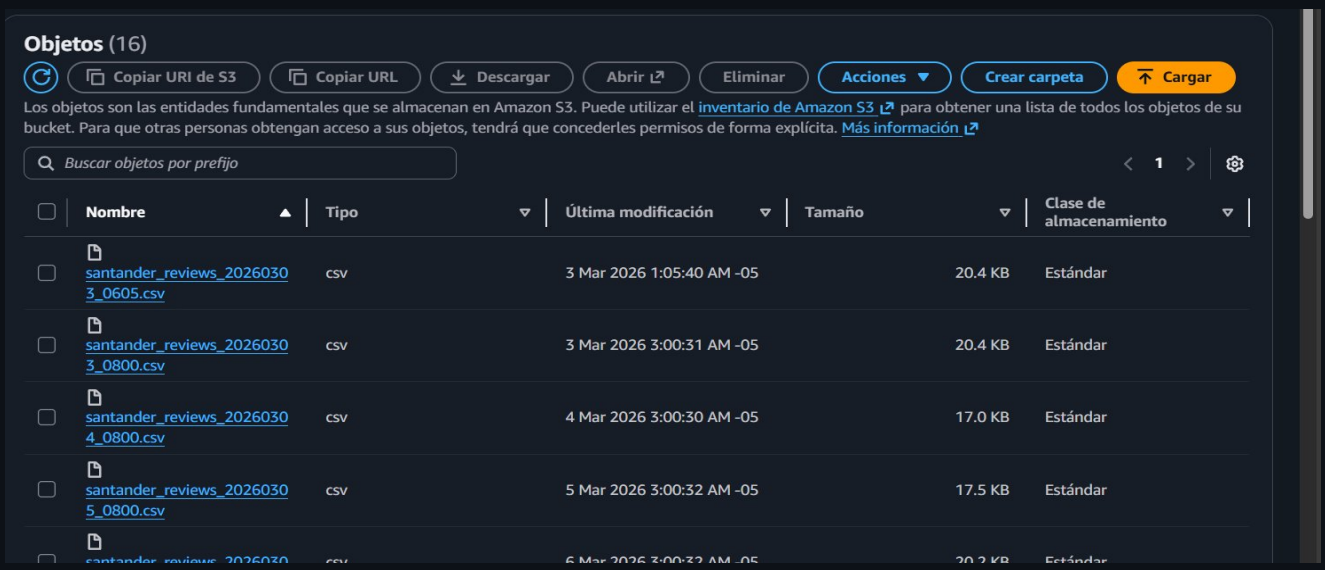
Layer	Technology	Role
Orchestration	AWS EventBridge + Lambda	Cron trigger · Serverless compute
Extraction	Python · Requests · BS4	Web scraping · HTML parsing
Transformation	Pandas · NLP Rules	Cleaning · Sentiment classification
Data Lake	Amazon S3	Raw backup · CSV archive
Data Warehouse	PostgreSQL / Supabase	Structured storage · SQL queries
Connectivity	SQLAlchemy + pg8000	Serverless-safe DB driver
Visualization	Microsoft Power BI	DirectQuery dashboards

05 · LIVE IMPLEMENTATION



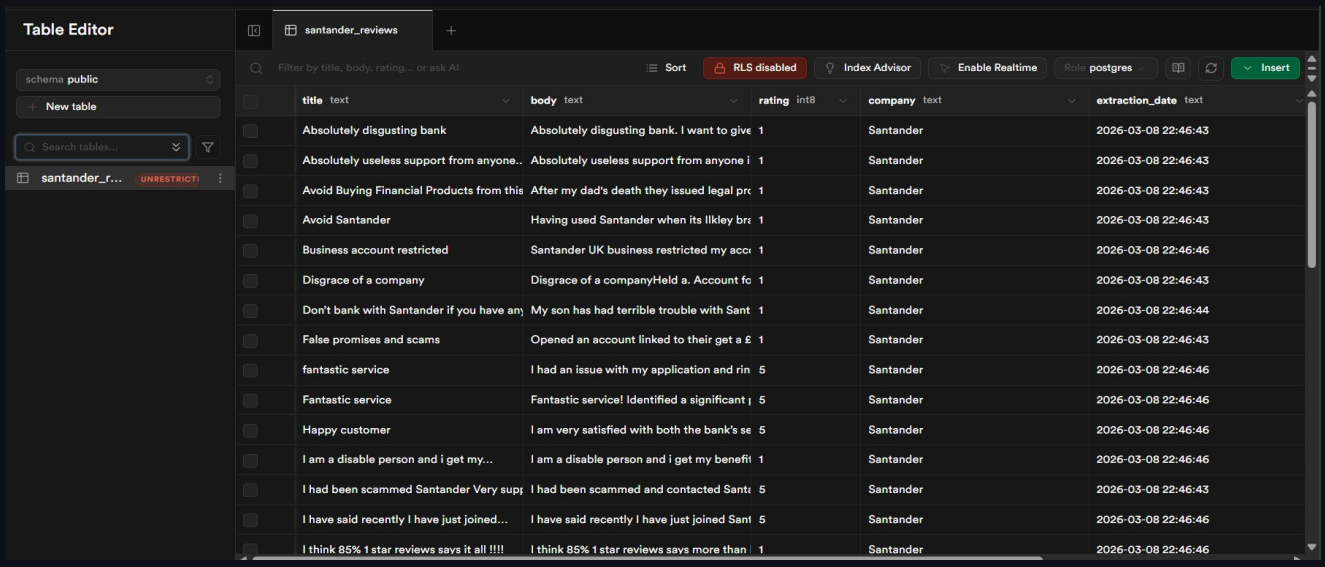
AWS Lambda — CustomerAnalyticsScraper function deployed with EventBridge trigger and one custom Layer for Python dependencies.

Amazon S3 Bucket — Root view showing the raw_data folder structure and the first CSV file automatically generated by the pipeline.



S3 Data Lake — 16 CSV files accumulated automatically, one per daily execution. Proof the cron job runs successfully every day.

EventBridge — BotScraperDiario rule in ENABLED state, confirming the scheduled cron trigger is active and operational.



Supabase PostgreSQL — santander_reviews table with real extracted data: title, body, rating, company and extraction timestamp.

Power BI Dashboard — Live sentiment analysis: 40 reviews, 75% negative, avg rating 1.95. Visualizations update daily from the warehouse.